

Recommending Green Routes for Pedestrians to Reduce the Exposure to Air Pollutants in Barcelona

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Pedestrians are Heavily Impacted by City Air Pollution



- *Poor air quality* in urban areas can lead *respiratory* and *cardiovascular diseases*
- *Pedestrians* are the *most exposed* individuals among those who commute in a city
- Can we *reduce the exposure* to air pollutants by walking *greener* routes?

Potential Users of Our Green Routes Prototype



Can I avoid that very polluted road,
even if I have to take a little detour?



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can we do better for our citizens?

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Main Features of Our Green Routes Prototype

- 👉 We recommend routes that *minimize* the *exposure* to air pollutants
- 📊 We consider *historical* Air Quality Index (AQI) data by Barcelona's city council
- 📶 We consider *real-time* AQI data from Barcelona's sensors (only 7 in the city)
- 🔗 We account for low-resolution data with our *GNN algorithm* (Calo *et al.*, 2024)

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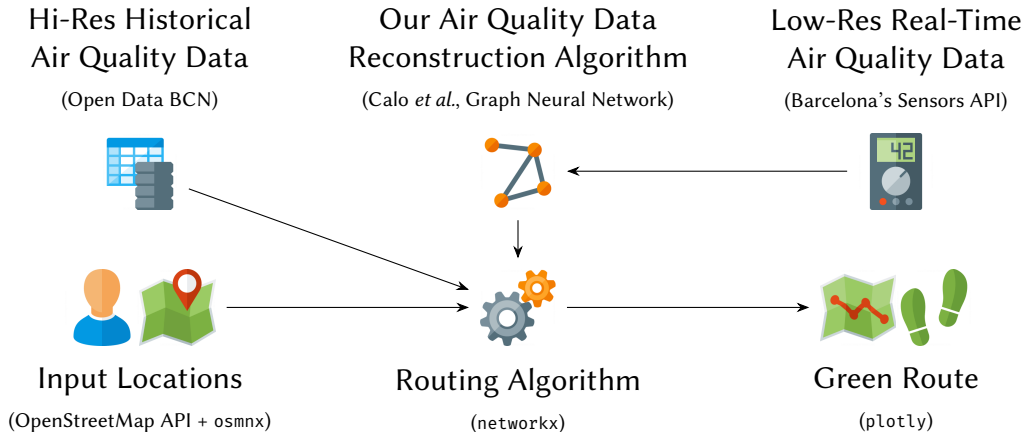
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System Architecture



Takeaway Message: Green Routes are Good!



- Tests with *Barcelona*'s NO₂ concentration data result in a median reduction of *-7.82%*
- *Encouraging* results have also been obtained in *London*, *California*, and *Melbourne*
- *Green routes* can be a simple, yet very effective solution to *improve citizens' health*